# Lecture 6: Four Fundamental Subspaces

MATH203 Applied Linear Algebra

# Motivation: Connecting the Subspaces

# Recall: The Preview from Lecture 5

At the end of Lecture 5, we discovered something beautiful:

**Theorem (Preview)**

$\text{Null}(A) \perp \text{Row}(A)$

Every vector in $\text{Null}(A)$ is orthogonal to every row of $A$.

**Why?** If $\mathbf{x} \in \text{Null}(A)$, then $A\mathbf{x} = \mathbf{0}$, which means:

$$\text{row}_i \cdot \mathbf{x} = 0 \quad \text{for all rows } i$$

**Today's question**: Is this an isolated phenomenon, or part of a bigger picture?

# What We'll Discover Today

For any matrix $A$, there are **four fundamental subspaces**:

| Subspace | Lives in | Dimension |
|---|---|---|
| $\text{Col}(A)$ | $\mathbb{R}^m$ (codomain) | $r$ |
| $\text{Null}(A)$ | $\mathbb{R}^n$ (domain) | $n - r$ |
| $\text{Col}(A^T)$ | $\mathbb{R}^n$ (domain) | $r$ |
| $\text{Null}(A^T)$ | $\mathbb{R}^m$ (codomain) | $m - r$ |

**Discovery**: They pair up into **orthogonal complements**!

# Roadmap for Today

**Strategy**: Build tools first, then reveal the full picture.

**Part 1**: How do subspaces behave under matrix products? ($AB$)

**Part 2**: What is transpose? (Flipping the ingredient table)

**Part 3**: The four fundamental subspaces and their orthogonality

**Part 4**: Dimension relationships

**Goal**: Understand the complete geometric structure of a matrix.

# Subspaces of Matrix Products

# Setup: Two-Stage Production

Imagine a **two-stage coffee shop**:

$$\begin{tikzcd}
\boxed{\text{Raw materials}} \arrow[r, "B"] \arrow[rr, bend left, "AB"] & \boxed{\text{Semi-products}} \arrow[r, "A"] & \boxed{\text{Final products}}
\end{tikzcd}$$

- $B$: raw materials $\to$ semi-products (e.g., roasted beans, brewed tea)
- $A$: semi-products $\to$ final products (e.g., latte, bubble tea)
- $AB$: raw materials $\to$ final products (combined process)

**Question**: How do the column spaces and null spaces of $A$, $B$, and $AB$ relate?

# Concrete Example: Two Matrices

Let's use actual matrices:

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \quad (3 \times 3)$$

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad (2 \times 3)$$

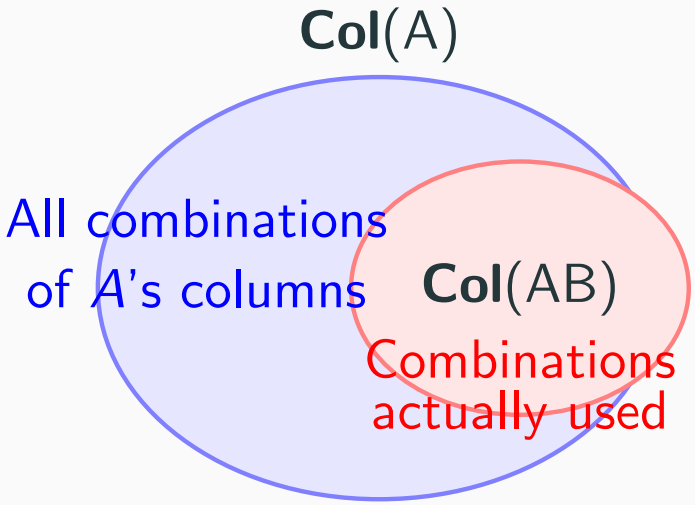
**Product** $AB$:

$$AB = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

We'll investigate: What are $\mathrm{Col}(A)$, $\mathrm{Col}(B)$, $\mathrm{Col}(AB)$?

# Visual: Column Space Containment

**Key observation**: Every column of $AB$ is built from columns of $A$.




**Why?** Column $j$ of $AB$ equals $A$ times column $j$ of $B$.

So every column of $AB$ is a **linear combination of $A$'s columns**.

# Theorem 1: Column Space of Product

**Theorem**

$\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$

The column space of $AB$ is always contained in the column space of $A$.

**Proof**:

Write $B$ with columns $\mathbf{b}_1, \ldots, \mathbf{b}_n$:

$$B = \begin{pmatrix} | & | & & | \\ \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_n \\ | & | & & | \end{pmatrix}$$

Then $AB$ has columns:

$$AB = \begin{pmatrix} | & | & & | \\ A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_n \\ | & | & & | \end{pmatrix}$$

Each $A\mathbf{b}_j$ is a linear combination of $A$'s columns. $\square$

# Why This Makes Sense: Column View of Matrix Product

Recall from Lecture 1: column $j$ of $AB$ equals $A\cdot$ (column $j$ of $B$).

**Example**: If $\mathbf{b}_j = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$ and $A$ has columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$:

$$A\mathbf{b}_j = 2\mathbf{a}_1 - 1\mathbf{a}_2 + 3\mathbf{a}_3$$

This is a **recipe** using $A$'s columns as ingredients!

**Conclusion**: Every column of $AB$ is cooked from $A$'s columns.

Therefore $\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$.

# Verify with Our Example

$$A = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad AB = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

**Column space of $AB$:**

$$\mathrm{Col}(AB) = \mathrm{span}\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \end{pmatrix} \right\}$$

Notice: $\begin{pmatrix} 5 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ (dependent!)

So actually: $\mathrm{Col}(AB) = \mathrm{span}\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \mathbb{R}^2$

**Column space of $A$:**

$$\mathrm{Col}(A) = \mathrm{span}\left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\} = \mathbb{R}^2$$



Indeed: $\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$ ✓ (Equality in this case!)

# Theorem 2: Null Space Inclusion

**Theorem**

$\text{Null}(B) \subseteq \text{Null}(AB)$

The null space of $B$ is always contained in the null space of $AB$.

**Proof**:

If $\mathbf{x} \in \text{Null}(B)$, then $B\mathbf{x} = \mathbf{0}$.

Therefore:

$$(AB)\mathbf{x} = A(B\mathbf{x}) = A(\mathbf{0}) = \mathbf{0}$$

So $\mathbf{x} \in \text{Null}(AB)$. $\square$

# Why This Makes Sense: Linear Dependencies Persist

Recall from Lecture 5: $\mathbf{x} \in \text{Null}(B)$ records a **linear dependence** among columns of $B$.

**Example**: If $\mathbf{x} = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix} \in \text{Null}(B)$, this says:

$$2\mathbf{b}_1 - 1\mathbf{b}_2 + 0\mathbf{b}_3 = \mathbf{0}$$

**What happens when we apply $A$ to both sides?**

$$A(2\mathbf{b}_1 - 1\mathbf{b}_2) = 2(A\mathbf{b}_1) - 1(A\mathbf{b}_2) = 2(AB)_1 - 1(AB)_2 = \mathbf{0}$$

**Conclusion**: The **same linear dependence** holds for columns of $AB$!

Applying $A$ preserves all existing linear relations.

# Verify with Our Example

$$B = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

**Find Null($B$)**:

By inspection, column 3 of $B$ equals $2\times$ column 1 plus $1\times$ column 2:

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 1 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

So the dependence relation is: $-2\mathbf{b}_1 - 1\mathbf{b}_2 + 1\mathbf{b}_3 = \mathbf{0}$

$$\text{Null}(B) = \text{span}\left\{ \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \right\}$$

# Verify with Our Example (continued)

$$AB = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix}$$

**Check**: Does $\begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \in \text{Null}(AB)$?

$$AB \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 & 5 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 - 1 + 5 \\ -2 + 0 + 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Yes! ✓

**Find Null($AB$) completely**:

Column 3 of $AB$ equals $2\times$ column 1 plus $1\times$ column 2:

$$\text{Null}(AB) = \text{span}\left\{ \begin{pmatrix} -2 \\ -1 \\ 1 \end{pmatrix} \right\}$$

# The Big Question

We've seen:

- $\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$ (always)
- $\mathrm{Null}(B) \subseteq \mathrm{Null}(AB)$ (always)

**But**: In our example, we had **equality** in both cases!

$$\textbf{When is Null}(AB) = \textbf{Null}(B) \textbf{ exactly?}$$

**Meaning**: When does multiplication by $A$ **not create new dependencies**?

# Theorem 3: The Equivalence

**Theorem**

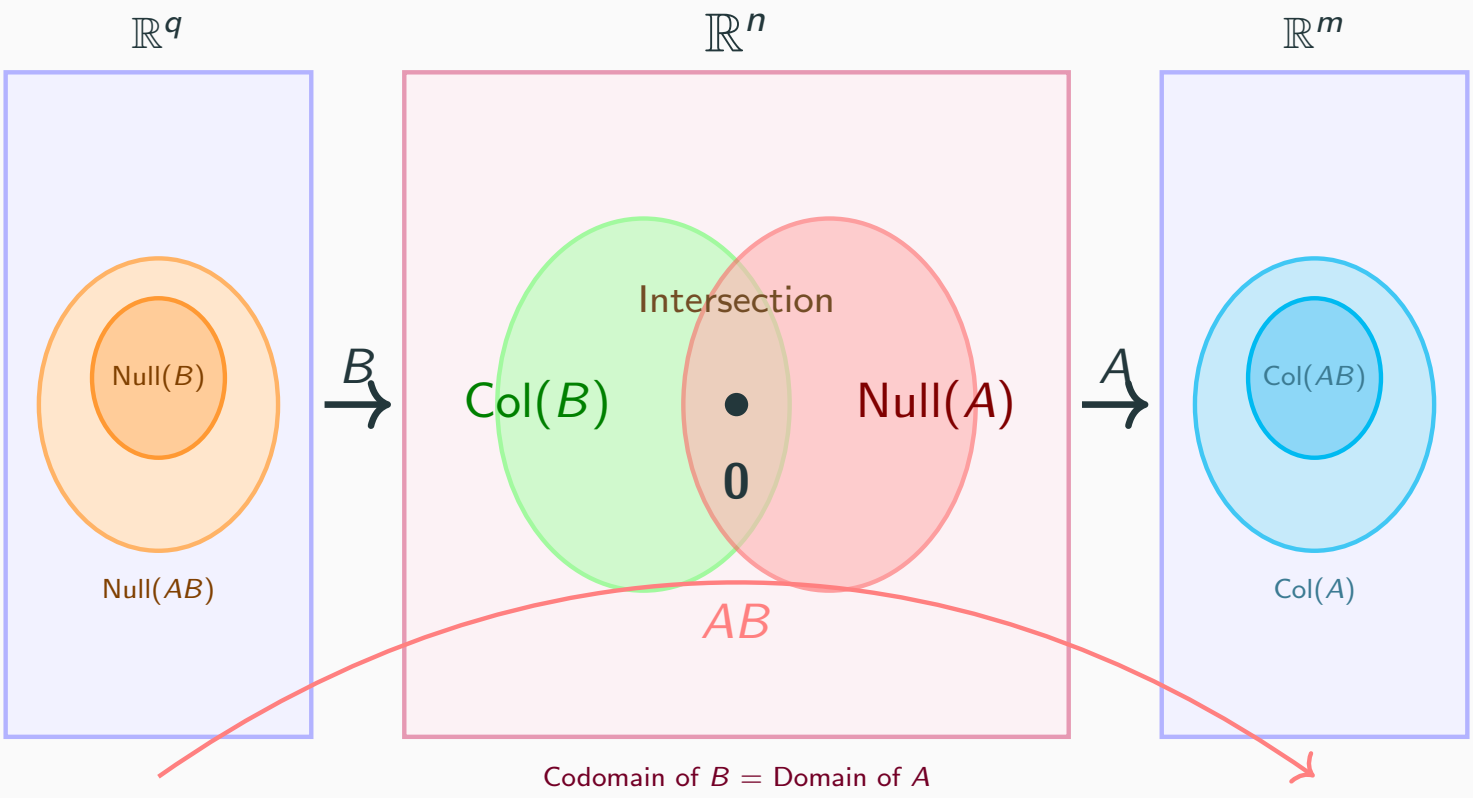
$$\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\} \quad \Longleftrightarrow \quad \text{Null}(AB) = \text{Null}(B)$$

**Left side**: Among all vectors $B$ can produce, none (except $\mathbf{0}$) lands in $\text{Null}(A)$.

**Right side**: All dependencies in $AB$ were already in $B$ (no new ones created).

These are **two perspectives on the same phenomenon**!

# Visual: The Equivalence




$$\text{Null}(AB) = \text{Null}(B) \iff \text{Col}(B) \cap \text{Null}(A) = \{\mathbf{0}\}$$

# Proof Direction (a): $\Rightarrow$

**Assume**: $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$

**Goal**: Show $\text{Null}(AB) = \text{Null}(B)$

We already know $\text{Null}(B) \subseteq \text{Null}(AB)$ (Theorem 2).

To show equality, prove the reverse inclusion.

Take any $\mathbf{x} \in \text{Null}(AB)$. Then $(AB)\mathbf{x} = \mathbf{0}$, so:

$$A(B\mathbf{x}) = \mathbf{0}$$

This means $B\mathbf{x} \in \text{Null}(A)$.

# Proof Direction (a): $\Rightarrow$ (continued)

We have: $B\mathbf{x} \in \text{Null}(A)$.

But also: $B\mathbf{x} \in \text{Col}(B)$ (by definition of column space).

Therefore: $B\mathbf{x} \in \text{Null}(A) \cap \text{Col}(B)$.

By assumption, $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$, so:

$$B\mathbf{x} = \mathbf{0}$$

This means $\mathbf{x} \in \text{Null}(B)$.

We've shown $\text{Null}(AB) \subseteq \text{Null}(B)$. Combined with Theorem 2:

$$\text{Null}(AB) = \text{Null}(B)$$

$\square$

# Proof Direction (b): $\Leftarrow$

**Assume**: $\text{Null}(AB) = \text{Null}(B)$

**Goal**: Show $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$

Suppose $\mathbf{v} \in \text{Null}(A) \cap \text{Col}(B)$.

Since $\mathbf{v} \in \text{Null}(A)$: $A\mathbf{v} = \mathbf{0}$.

Since $\mathbf{v} \in \text{Col}(B)$: $\mathbf{v} = B\mathbf{x}$ for some $\mathbf{x}$.

Combining:

$$A(B\mathbf{x}) = A\mathbf{v} = \mathbf{0}$$

So $(AB)\mathbf{x} = \mathbf{0}$, meaning $\mathbf{x} \in \text{Null}(AB)$.

# Proof Direction (b): $\Leftarrow$ (continued)

We have: $\mathbf{x} \in \text{Null}(AB)$.

By hypothesis, $\text{Null}(AB) = \text{Null}(B)$, so:

$$\mathbf{x} \in \text{Null}(B)$$

Therefore:

$$B\mathbf{x} = \mathbf{0}$$

But we had $\mathbf{v} = B\mathbf{x}$, so:

$$\mathbf{v} = \mathbf{0}$$

We've shown: any $\mathbf{v} \in \text{Null}(A) \cap \text{Col}(B)$ must be $\mathbf{0}$.

Therefore: $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\}$ $\square$

# Summary: Subspaces of Matrix Products

| **Theorem** | **Meaning** |
|---|---|
| $\mathrm{Col}(AB) \subseteq \mathrm{Col}(A)$ | Columns of $AB$ built from $A$'s columns |
| $\mathrm{Null}(B) \subseteq \mathrm{Null}(AB)$ | Dependencies in $B$ persist in $AB$ |
| $\mathrm{Null}(A) \cap \mathrm{Col}(B) = \{\mathbf{0}\}$ $\Leftrightarrow$ $\mathrm{Null}(AB) = \mathrm{Null}(B)$ | No new dependencies created |

**Key insight**: Matrix multiplication has predictable effects on subspaces.

**Next**: We need a new tool to reveal deeper relationships — **transpose**.

# Transpose

# What Is Transpose?

The **transpose** of a matrix is obtained by **flipping it along the main diagonal**.

**Effect**: Rows become columns, and columns become rows.

**Notation**: For matrix $A$, its transpose is written $A^T$.

**Example**:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \xrightarrow{\text{transpose}} A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Row 1 of $A$: $(1, 2, 3)$ becomes column 1 of $A^T$

# Visual: Flipping Along the Diagonal

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \xrightarrow{\text{Flip along diagonal}} \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

$A$ $\qquad$ $A^T$

**Key**: Entry $(i, j)$ of $A$ becomes entry $(j, i)$ of $A^T$.

# More Examples

**Example 1**: Square matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \Rightarrow \quad A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

**Example 2**: Tall matrix $\rightarrow$ wide matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \quad \Rightarrow \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

**Example 3**: Column vector $\rightarrow$ row vector

$$A = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \quad \Rightarrow \quad A^T = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$$

# Formal Definition

**Definition (Transpose)**

For an $m \times n$ matrix $A$, its **transpose** $A^T$ is the $n \times m$ matrix:

$$(A^T)_{ij} = A_{ji}$$

Rows of $A$ become columns of $A^T$, and columns of $A$ become rows of $A^T$.

**Example**:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad \Rightarrow \quad A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}$$

Row 1 of $A$: $(1, 2, 3)$ becomes column 1 of $A^T$: $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

# Property 1: Transpose of Product

**Proposition**

$$(AB)^T = B^T A^T$$

The transpose of a product reverses the order!

**Proof** (entry-by-entry):

Entry $(i, j)$ of $(AB)^T$:

$$(AB)^T_{ij} = (AB)_{ji} = \sum_k A_{jk} B_{ki}$$

Entry $(i, j)$ of $B^T A^T$:

$$(B^T A^T)_{ij} = \sum_k (B^T)_{ik} (A^T)_{kj} = \sum_k B_{ki} A_{jk}$$

Since multiplication of numbers commutes: $\sum_k A_{jk} B_{ki} = \sum_k B_{ki} A_{jk}$ $\square$

# Property 2: Transpose of Inverse

**Proposition**

If $A$ is invertible:

$$(A^{-1})^T = (A^T)^{-1}$$

Transpose and inverse commute.

**Proof**:

Start from $A \cdot A^{-1} = I$, transpose both sides:

$$(A \cdot A^{-1})^T = I^T = I$$

Using Property 1:

$$(A^{-1})^T \cdot A^T = I$$

Similarly, $(A^{-1} \cdot A)^T = I$ gives $A^T \cdot (A^{-1})^T = I$.

Therefore $(A^{-1})^T$ is the inverse of $A^T$. $\square$

# Property 3: Transpose is Linear

**Proposition**

$$(A+B)^T = A^T + B^T$$

Transpose distributes over addition.

**Proof** (entry-by-entry):

$$\begin{aligned}(A+B)^T_{ij} &= (A+B)_{ji} \\ &= A_{ji} + B_{ji} \\ &= (A^T)_{ij} + (B^T)_{ij} \\ &= (A^T + B^T)_{ij}\end{aligned}$$

Therefore $(A+B)^T = A^T + B^T$. $\square$

# Symmetric Matrices

**Definition (Symmetric Matrix)**

A matrix $S$ is **symmetric** if $S^T = S$.

The matrix looks the same when transposed!

**Example**:

$$S = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$$

Flipping rows $\leftrightarrow$ columns gives the same matrix.

**Notice**: $S_{ij} = S_{ji}$ for all $i, j$ (mirror symmetry across diagonal).

# Theorem: $AA^T$ and $A^TA$ Are Always Symmetric

**Theorem**

For any matrix $A$:

1. $AA^T$ is symmetric
2. $A^TA$ is symmetric

**Proof**:

Using $(AB)^T = B^TA^T$:

$$(AA^T)^T = (A^T)^TA^T = AA^T$$

So $AA^T$ is symmetric.

Similarly:

$$(A^TA)^T = A^T(A^T)^T = A^TA$$

So $A^TA$ is symmetric. $\square$

# Verify with Concrete Matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

**Compute $AA^T$:**

$$AA^T = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} = \begin{pmatrix} 5 & 11 & 17 \\ 11 & 25 & 39 \\ 17 & 39 & 61 \end{pmatrix}$$

Check: $(AA^T)^T = AA^T$ ✓ (symmetric!)

# Verify with Concrete Matrix (continued)

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix}, \quad A^T = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix}$$

**Compute $A^T A$:**

$$A^T A = \begin{pmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} = \begin{pmatrix} 35 & 44 \\ 44 & 56 \end{pmatrix}$$

Check: $(A^T A)^T = A^T A$ ✓ (symmetric!)

# Summary: Transpose Properties

**What we've established:**

- **Definition**: $A^T$ is obtained by flipping $A$ along the main diagonal
- **Product rule**: $(AB)^T = B^T A^T$ (order reverses)
- **Inverse rule**: $(A^{-1})^T = (A^T)^{-1}$
- **Linearity**: $(A + B)^T = A^T + B^T$
- **Symmetric matrices**: $AA^T$ and $A^T A$ are always symmetric

These properties apply to **all matrices**, and we'll use them to discover deeper relationships between the four fundamental subspaces.

**Next**: Orthogonality relationships between subspaces.

# Four Fundamental Subspaces and Orthogonality

# The Four Fundamental Subspaces

For any $m \times n$ matrix $A$, there are **four fundamental subspaces**:

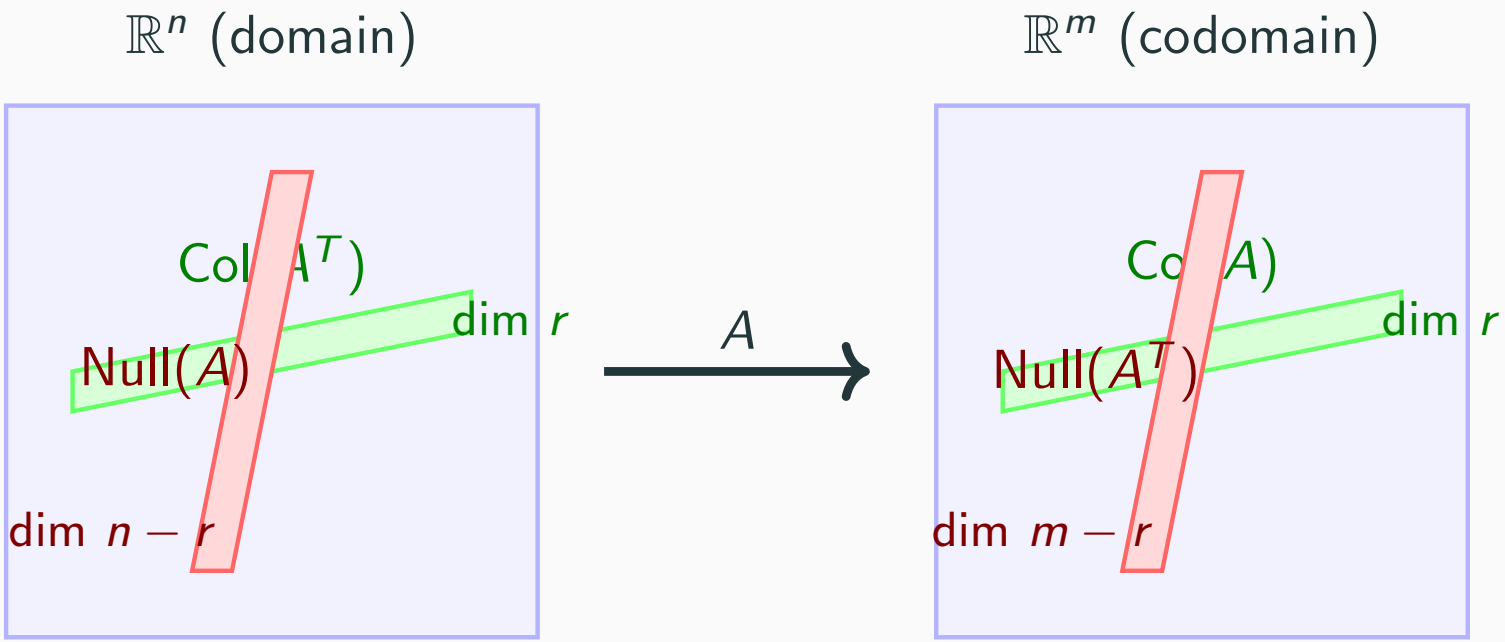
| Subspace | Definition | Lives in | Dimension |
|---|---|---|---|
| $\text{Col}(A)$ | $\{A\mathbf{x}\}$ | $\mathbb{R}^m$ | $r$ |
| $\text{Null}(A)$ | $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$ | $\mathbb{R}^n$ | $n - r$ |
| $\text{Col}(A^T)$ | $\{A^T\mathbf{y}\}$ | $\mathbb{R}^n$ | $r$ |
| $\text{Null}(A^T)$ | $\{\mathbf{y} : A^T\mathbf{y} = \mathbf{0}\}$ | $\mathbb{R}^m$ | $m - r$ |

where $r = \text{rank}(A)$.

**Key observations**:

- Two subspaces in $\mathbb{R}^n$ (domain): $\text{Null}(A)$ and $\text{Col}(A^T)$
- Two subspaces in $\mathbb{R}^m$ (codomain): $\text{Col}(A)$ and $\text{Null}(A^T)$

# Visual: Two Pairs of Subspaces




**Today's discovery**: These pairs are **orthogonal** to each other!

# Theorem 4: $\mathbf{Col}(A^T) \perp \mathbf{Null}(A)$

**Theorem**

Every vector in $\text{Col}(A^T)$ is orthogonal to every vector in $\text{Null}(A)$.

Precisely: For $\mathbf{x} \in \text{Null}(A)$ and $\mathbf{y} \in \text{Col}(A^T)$,

$$\mathbf{y}^T\mathbf{x} = 0$$

**Meaning**: The row space and null space are perpendicular!

# Proof (Inner Product View)

Let $\mathbf{x} \in \text{Null}(A)$ and $\mathbf{y} \in \text{Col}(A^T)$.

By definition:

- $\mathbf{x} \in \text{Null}(A)$ means $A\mathbf{x} = \mathbf{0}$
- $\mathbf{y} \in \text{Col}(A^T)$ means $\mathbf{y} = A^T\mathbf{z}$ for some $\mathbf{z}$

Compute the inner product:

$$\begin{aligned} \mathbf{y}^T\mathbf{x} &= (A^T\mathbf{z})^T\mathbf{x} \\ &= \mathbf{z}^T(A^T)^T\mathbf{x} \\ &= \mathbf{z}^T A\mathbf{x} \end{aligned}$$

Since $\mathbf{x} \in \text{Null}(A)$, we have $A\mathbf{x} = \mathbf{0}$:

$$\mathbf{z}^T A\mathbf{x} = \mathbf{z}^T\mathbf{0} = 0$$

$\square$

# Proof (Geometric View: Rows as Normal Vectors)

The equation $A\mathbf{x} = \mathbf{0}$ can be written row-by-row:

$$\begin{pmatrix} —\mathbf{r}_1^T— \\ —\mathbf{r}_2^T— \\ \vdots \\ —\mathbf{r}_m^T— \end{pmatrix} \mathbf{x} = \begin{pmatrix} \mathbf{r}_1^T\mathbf{x} \\ \mathbf{r}_2^T\mathbf{x} \\ \vdots \\ \mathbf{r}_m^T\mathbf{x} \end{pmatrix} = \mathbf{0}$$

This means:

$$\mathbf{r}_1^T\mathbf{x} = 0, \quad \mathbf{r}_2^T\mathbf{x} = 0, \quad \ldots, \quad \mathbf{r}_m^T\mathbf{x} = 0$$

**Interpretation**: Each row $\mathbf{r}_i$ is **perpendicular** to $\mathbf{x}$.

# Proof (Geometric View: continued)

We have: Each row $\mathbf{r}_i$ of $A$ is perpendicular to every $\mathbf{x} \in \text{Null}(A)$.

**But**: The rows of $A$ are exactly the **columns of $A^T$**!

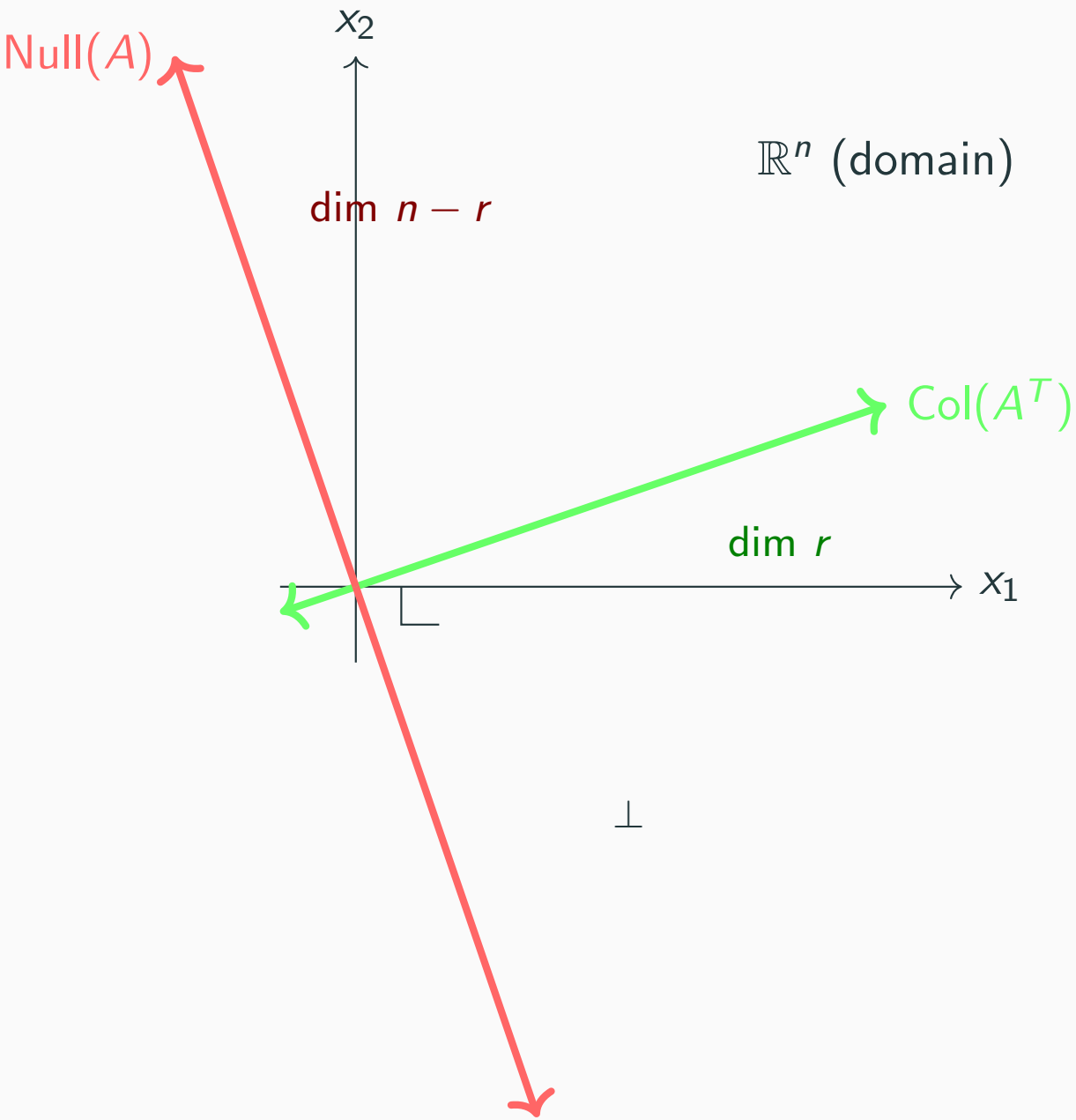
$$A^T = \begin{pmatrix} | & | & & | \\ \mathbf{r}_1 & \mathbf{r}_2 & \cdots & \mathbf{r}_m \\ | & | & & | \end{pmatrix}$$

Therefore:

$$\text{Col}(A^T) = \text{span}\{\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_m\}$$

consists of vectors perpendicular to $\text{Null}(A)$. $\square$

# Visual: Orthogonality in $\mathbb{R}^n$



# Concrete Example

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$$

**Find Null($A$)**:

$A\mathbf{x} = \mathbf{0}$ gives: $x_1 + 2x_2 + 3x_3 = 0$ (second row is $2\times$ first)

Basis for Null($A$):

$$\text{Null}(A) = \text{span}\left\{ \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ 0 \\ 1 \end{pmatrix} \right\}$$

**Find Col($A^T$)**:

$$A^T = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$$

# Verify Orthogonality

$$\mathrm{Col}(A^T) = \mathrm{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}\right\}, \quad \mathrm{Null}(A) = \mathrm{span}\left\{\begin{pmatrix}-2\\1\\0\end{pmatrix}, \begin{pmatrix}-3\\0\\1\end{pmatrix}\right\}$$

**Check orthogonality**:

$$\begin{pmatrix}1\\2\\3\end{pmatrix}^T \begin{pmatrix}-2\\1\\0\end{pmatrix} = 1(-2) + 2(1) + 3(0) = 0 \quad \checkmark$$

$$\begin{pmatrix}1\\2\\3\end{pmatrix}^T \begin{pmatrix}-3\\0\\1\end{pmatrix} = 1(-3) + 2(0) + 3(1) = 0 \quad \checkmark$$

Indeed, $\mathrm{Col}(A^T) \perp \mathrm{Null}(A)$!

# Theorem 5: $\mathbf{Col}(A) \perp \mathbf{Null}(A^T)$

**Theorem**

Every vector in $\text{Col}(A)$ is orthogonal to every vector in $\text{Null}(A^T)$.

Precisely: For $\mathbf{x} \in \text{Col}(A)$ and $\mathbf{y} \in \text{Null}(A^T)$,

$$\mathbf{y}^T\mathbf{x} = 0$$

**Meaning**: The column space and left null space are perpendicular!

# Proof

Let $\mathbf{y} \in \text{Null}(A^T)$ and $\mathbf{x} \in \text{Col}(A)$.

By definition:

- $\mathbf{y} \in \text{Null}(A^T)$ means $A^T\mathbf{y} = \mathbf{0}$
- $\mathbf{x} \in \text{Col}(A)$ means $\mathbf{x} = A\mathbf{z}$ for some $\mathbf{z}$

Compute the inner product:

$$\begin{aligned} \mathbf{y}^T\mathbf{x} &= \mathbf{y}^T(A\mathbf{z}) \\ &= (A^T\mathbf{y})^T\mathbf{z} \end{aligned}$$

Since $A^T\mathbf{y} = \mathbf{0}$:

$$(A^T\mathbf{y})^T\mathbf{z} = \mathbf{0}^T\mathbf{z} = 0$$

$\square$

# Symmetry of the Theorems

**Beautiful duality**:

- **Theorem 4**: $\text{Col}(A^T) \perp \text{Null}(A)$ (both in $\mathbb{R}^n$)
- **Theorem 5**: $\text{Col}(A) \perp \text{Null}(A^T)$ (both in $\mathbb{R}^m$)

These are **not two separate facts** — they're the **same theorem**!

**Why?** Apply Theorem 4 to the matrix $A^T$ instead of $A$:

$$\text{Col}((A^T)^T) \perp \text{Null}(A^T) \quad \Rightarrow \quad \text{Col}(A) \perp \text{Null}(A^T)$$

which is exactly Theorem 5!

**Deep insight**: Transpose reveals a fundamental symmetry in linear algebra.

# Concrete Example: The Dual Case

Using the same matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$$

**Find Col($A$)**:

$$\text{Col}(A) = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 6 \end{pmatrix} \right\} = \text{span}\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$$

**Find Null($A^T$)**:

$A^T\mathbf{y} = \mathbf{0}$ gives: $\begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \mathbf{0}$

All equations reduce to: $y_1 + 2y_2 = 0$

$$\text{Null}(A^T) = \text{span}\left\{ \begin{pmatrix} -2 \\ 1 \end{pmatrix} \right\}$$

# Concrete Example: Verify Orthogonality

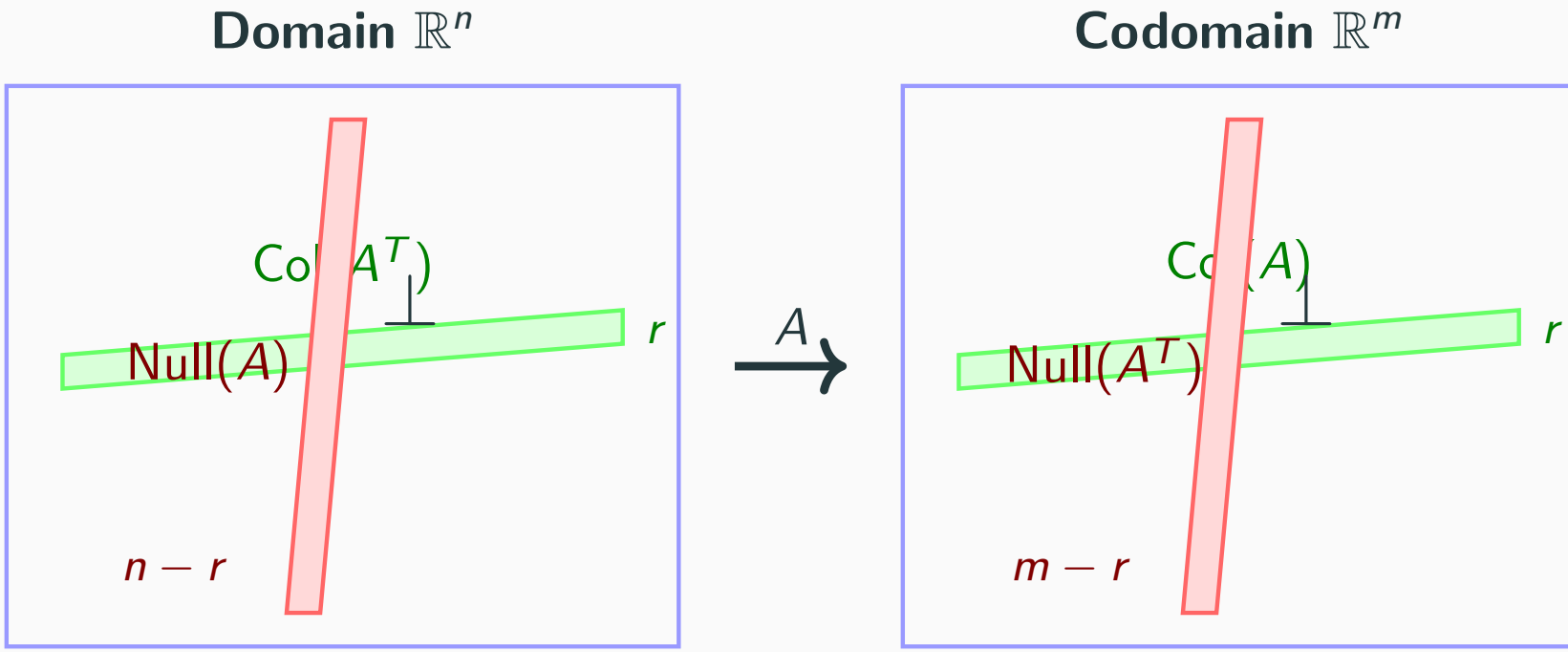
$$\mathrm{Col}(A) = \mathrm{span}\left\{\begin{pmatrix}1\\2\end{pmatrix}\right\}, \quad \mathrm{Null}(A^T) = \mathrm{span}\left\{\begin{pmatrix}-2\\1\end{pmatrix}\right\}$$

**Check orthogonality**:

$$\begin{pmatrix}-2\\1\end{pmatrix}^T \begin{pmatrix}1\\2\end{pmatrix} = (-2)(1) + (1)(2) = -2 + 2 = 0 \quad \checkmark$$

Indeed, $\mathrm{Col}(A) \perp \mathrm{Null}(A^T)$!

# The Big Picture: Four Subspaces




$$\dim \mathrm{Col}(A^T) + \dim \mathrm{Null}(A) = r + (n - r) = n \qquad \dim \mathrm{Col}(A) + \dim \mathrm{Null}(A^T) = r + (m$$

**Key**: The orthogonal pairs completely fill their spaces!

# Dimension Relationships

# Theorem 6: Row Rank Equals Column Rank

**Theorem**

$$\text{rank}(A) = \text{rank}(A^T)$$

The dimension of $\text{Col}(A)$ equals the dimension of $\text{Col}(A^T)$.

In other words: **Column rank = Row rank**.

**Proof idea** (using cross-filling from Lecture 3):

Any matrix can be decomposed:

$$A = R_1 + R_2 + \cdots + R_r = UV$$

where $U$ is $m \times r$ and $V$ is $r \times n$.

# Proof of Theorem 6 (continued)

$$A = UV$$

where:

- $U$ is $m \times r$ with $r$ linearly independent columns
- $V$ is $r \times n$ with $r$ linearly independent rows

**Column rank**:

$$\mathrm{Col}(A) = \mathrm{Col}(UV) = \mathrm{Col}(U)$$

So $\dim(\mathrm{Col}(A)) = r$.

**Row rank**:

The rows of $A = UV$ are linear combinations of the rows of $V$.

Since $V$ has $r$ linearly independent rows, $\dim(\mathrm{Col}(A^T)) = r$.

Therefore $\mathrm{rank}(A) = \mathrm{rank}(A^T) = r$. $\square$

# Dimension Formulas

From the rank-nullity theorem (Lecture 5) and orthogonality:

**Proposition**

For an $m \times n$ matrix $A$ with $\text{rank}(A) = r$:

**In $\mathbb{R}^n$ (domain):**

$$\dim(\text{Col}(A^T)) + \dim(\text{Null}(A)) = r + (n - r) = n$$

**In $\mathbb{R}^m$ (codomain):**

$$\dim(\text{Col}(A)) + \dim(\text{Null}(A^T)) = r + (m - r) = m$$

**Meaning**: The orthogonal pairs together **span their entire spaces**.

# Summary

# What We Achieved Today

1. **Subspaces of Matrix Products**:
   - $\text{Col}(AB) \subseteq \text{Col}(A)$ — columns of $AB$ built from $A$'s columns
   - $\text{Null}(B) \subseteq \text{Null}(AB)$ — dependencies persist
   - $\text{Null}(A) \cap \text{Col}(B) = \{\mathbf{0}\} \Leftrightarrow \text{Null}(AB) = \text{Null}(B)$ — no new dependencies

2. **Transpose**:
   - Flipping the ingredient table perspective
   - $(AB)^T = B^T A^T$ — order reverses
   - $AA^T$ and $A^T A$ always symmetric

# What We Achieved Today (continued)

3. **Four Fundamental Subspaces**:

- $\text{Col}(A^T) \perp \text{Null}(A)$ in $\mathbb{R}^n$
- $\text{Col}(A) \perp \text{Null}(A^T)$ in $\mathbb{R}^m$
- These are dual statements — same theorem applied to $A$ and $A^T$

4. **Dimension Relationships**:

- $\text{rank}(A) = \text{rank}(A^T)$ — column rank equals row rank
- $\dim(\text{Col}(A^T)) + \dim(\text{Null}(A)) = n$
- $\dim(\text{Col}(A)) + \dim(\text{Null}(A^T)) = m$

**Key insight**: The four fundamental subspaces form two orthogonal pairs that completely describe the geometry of a matrix.

# Preview: Next Lecture

**Lecture 7**: Linear Transformations and Projections

Now that we understand the four fundamental subspaces and their orthogonality, we can study:

- What are linear transformations?
- What are projection operators? ($P^2 = P$)
- How do projections relate to the four subspaces?
- The geometric meaning of $\text{Col}(P)$ and $\text{Null}(P)$

**Goal**: Understand how to decompose vectors into orthogonal components.